Title: Tech Employment in the Age of AI
We Came, We Saw, We Modeled
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DS 4002
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Goal Statement: This project aims to forecast global monthly tech layoffs for the remainder of 2026 and 2027 using time series data from 2020-present; the model will succeed if it achieves a lower RMSE on a held-out 12-month test set than a seasonal naive baseline.

Research Question: To what extent has the development and adoption of AI systems impacted employment levels in the global technology sector, and what do time-series modeling of 2020-2025 suggest about the likelihood of recovery in the next few years?

Executive Summary: This project analyzes global technology-sector layoffs from 2020 to the present in order to forecast monthly layoff trends through 2027. For this milestone, we performed exploratory data analysis to identify key variables, examine temporal patterns in layoffs, and assess the statistical properties of the time series. EDA indicates the series is non-stationary with seasonal structure.

Data Set Summary: Our data set contains records of technology-sector layoffs collected from publicly reported layoff events from 2020 to the present. Each observation represents a reported layoff event and includes the company name, the date of the layoffs, the number of employees laid off, the company's industry, and additional company characteristics such as funding stage and amount of money raised. Variables related to geographic location (e.g., city, state, country, and continent) will not be included in our analysis because this project focuses on modeling overall time trends in technology-sector layoffs rather than comparing layoffs across regions.

Data Dictionary:

Variable Name	Type	Description	Role in Analysis
Nr	Integer	Unique identifier assigned to each layoff event in the dataset.	Identifier
Company	String	Name of the company that reported the layoff event.	Descriptive variable
Date_layoffs	Date	Date when the layoff event occurred or was publicly reported.	Primary time variable used to aggregate layoffs into a monthly time series
Laid_Off	Numeric	Number of employees laid off during the reported event.	Main outcome variable used to construct the layoff time series

Percentage	Numeric	Percentage of the company's workforce that was laid off during the event.	Contextual variable describing the scale of layoffs
Company_Size _before_Layoffs	Numeric	Estimated number of employees at the company before the layoffs occurred.	Contextual variable
Company_Size _after_layoffs	Numeric	Estimated number of employees remaining after the layoffs occurred.	Contextual variable
Industry	String	Industry category that the company belongs to (e.g., fintech, AI, e-commerce).	Descriptive / grouping variable
Stage	String	Business or funding stage of the company (e.g., startup, post-IPO).	Contextual variable
Money_Raised_ in__mil	Numeric	Total amount of funding the company raised, measured in millions of dollars.	Contextual variable
Year	Integer	Year in which the layoff event occurred.	Time variable used for exploratory analysis.

Questions Explored in EDA: Several exploratory questions were examined to better understand the data and determine whether it is suitable for time-series forecasting. Specifically, we explored how layoffs have changed over time, whether the series is stationary, whether layoffs show seasonal patterns, and what the ACF and PACF plots suggest about model order.

The first chart examines how layoffs have trended over time by aggregating the data into monthly totals. The time-series plot shows that layoffs increased in early 2020, likely reflecting the economic disruption caused by the COVID-19 pandemic. After this spike, layoffs remained relatively low through most of 2021. However, layoffs began to rise again in 2022 and increased dramatically in late 2022 and early 2023, with monthly layoffs exceeding 60,000 employees. These large fluctuations suggest that the series is not stationary in its original form.

The next chart looks at total layoffs by year. Layoffs were moderate in 2020 (about 61,500 employees) and very low in 2021 (around 6,800 employees), when many tech companies were still expanding. Layoffs then increased sharply in 2022 and peaked in 2023 with roughly 176,946 employees laid off. These numbers are much higher than during the COVID period and suggest a major restructuring phase in the tech industry as companies adjust to the rapid growth of AI technologies.

The heatmap shows that layoffs are not evenly distributed across months and tend to occur in clusters during certain periods of industry contraction. However, the pattern is not perfectly

consistent each year, suggesting layoffs are influenced more by broader economic conditions than strict seasonal cycles.

Finally, we examined the ACF and PACF plots to better understand the statistical structure of the series. The ACF decays slowly, indicating the original series is likely non-stationary. After differencing, the autocorrelation drops more quickly, suggesting that differencing helps remove the trend. The PACF shows strong spikes at early lags, indicating that layoffs in recent months are strong predictors of layoffs in the following months.

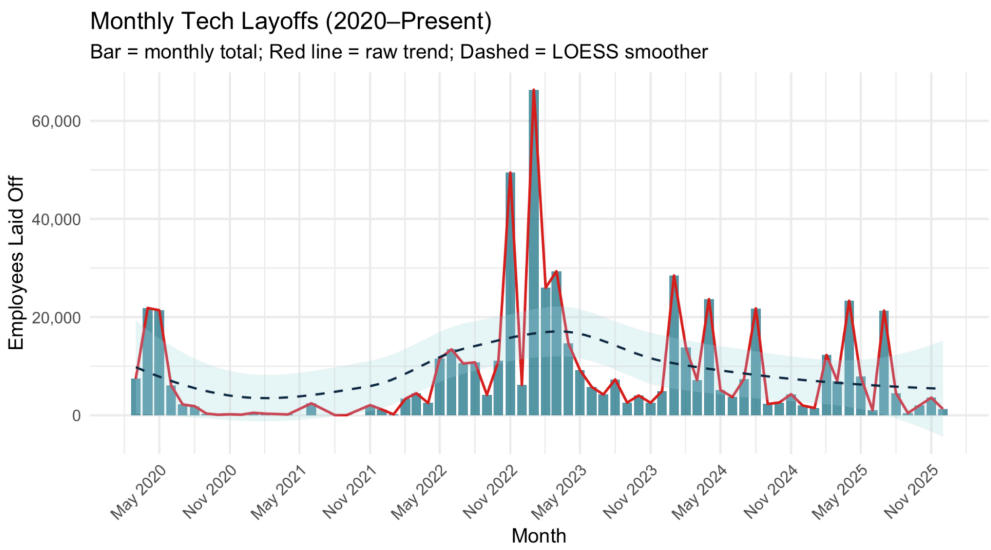
Current Unknowns or Questions: Although the dataset provides a large number of reported layoff events, there are a couple of uncertainties that could influence the analysis.

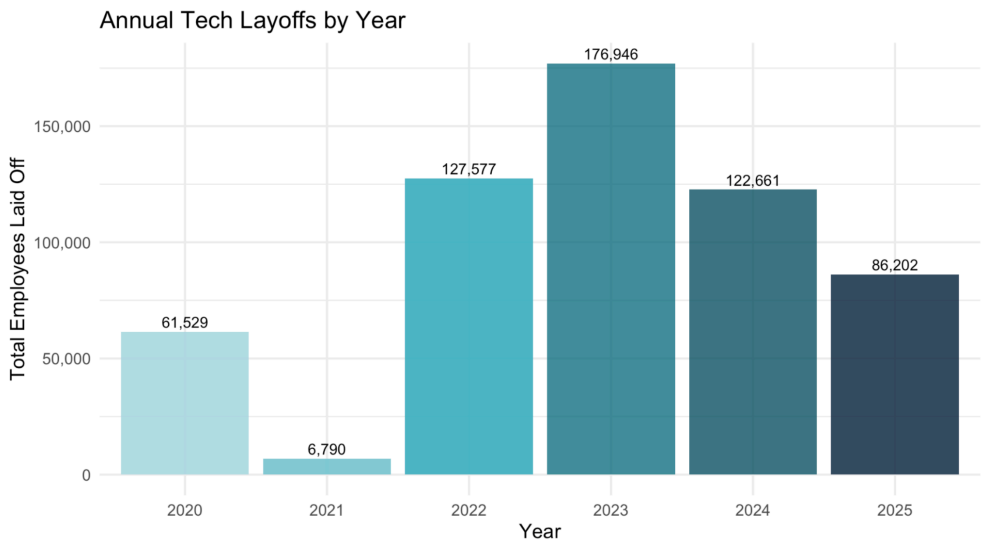
Data Completeness: The dataset relies on publicly reported layoffs and may underestimate events at smaller companies or in non-English-speaking markets. This limits generalizability of the global forecast but is acceptable given the US-dominant nature of the data.

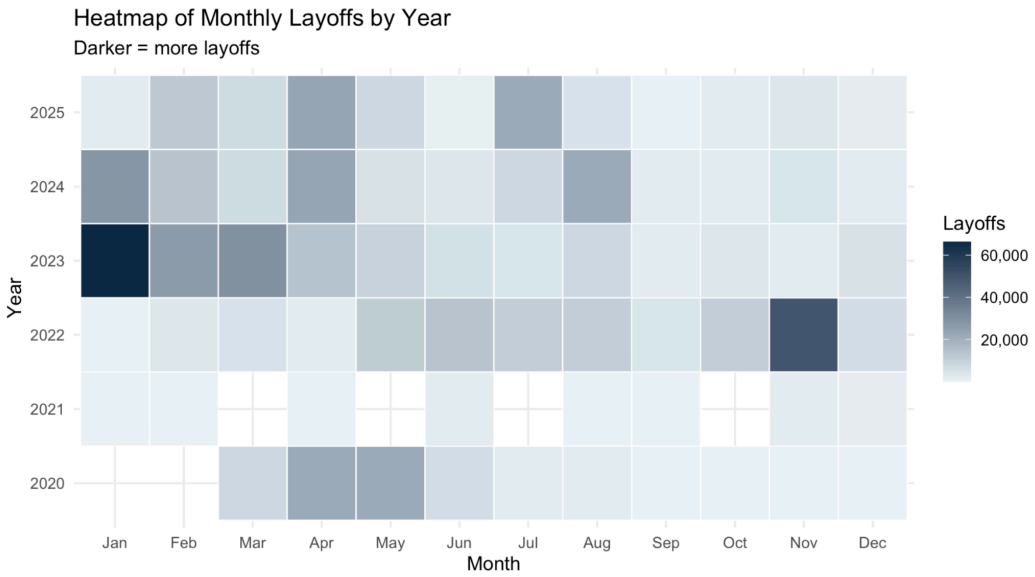
Parameter Selection Uncertainty: While ACF/PACF plots suggest starting candidates for model order, the final p, q, P, and Q values will be determined by auto.arima(), which systematically evaluates AIC and BIC across candidate models. This mitigates the risk of manual misspecification.

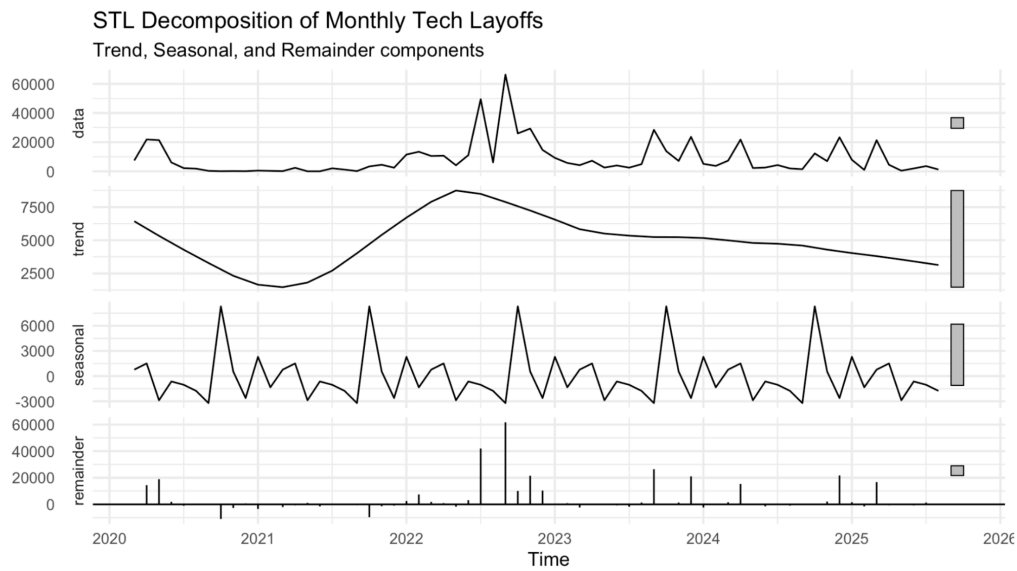
Refinement of Research Question/Model Plan: Despite uncertainties, the dataset still provides a sufficiently long time series of monthly layoff totals to support time-series modeling. The analysis plan addresses limitations through data cleaning, differencing to remove non-stationarity, and careful model selection using statistical criteria such as AIC, BIC, and RMSE. Based on the seasonality detected in the STL decomposition, the model has been refined from ARIMA to SARIMA to better capture the Q1 clustering pattern present across years. Additionally, given the pronounced spike in early 2023, we are evaluating whether the inclusion of a binary external regressor improves out-of-sample forecast accuracy. The research question itself remains unchanged, but the modeling plan has been updated to evaluate SARIMA(1,1,1)(1,1,0)[12] and ARIMA(2,1,1) as primary candidates, with auto.arima() used for systematic parameter selection.

Exploratory Plots: Figure 1 shows monthly tech layoffs from May 2020 through November 2025. The bar chart displays monthly totals and the LOESS smoother (dashed) reveals the overall trend arc, a plateau through 2020-2021, a sharp rise in 2022, a peak in early 2023, and a partial recovery through 2025. Figure 2 presents annual totals, illustrating the scale contrast between 2021 (6,790 employees) and 2023 (176,946 employees). Figure 3 is a year-by-month heatmap that reveals the Q1 seasonal clustering pattern across years- the darkest cells appear in January-March of 2023 and 2024, confirming the seasonal signal identified in the STL decomposition. The STL decomposition (Figure 4) and ACF/PACF plots (Figure 5) are included as direct justification for the SARIMA modeling approach and parameter selection strategy.

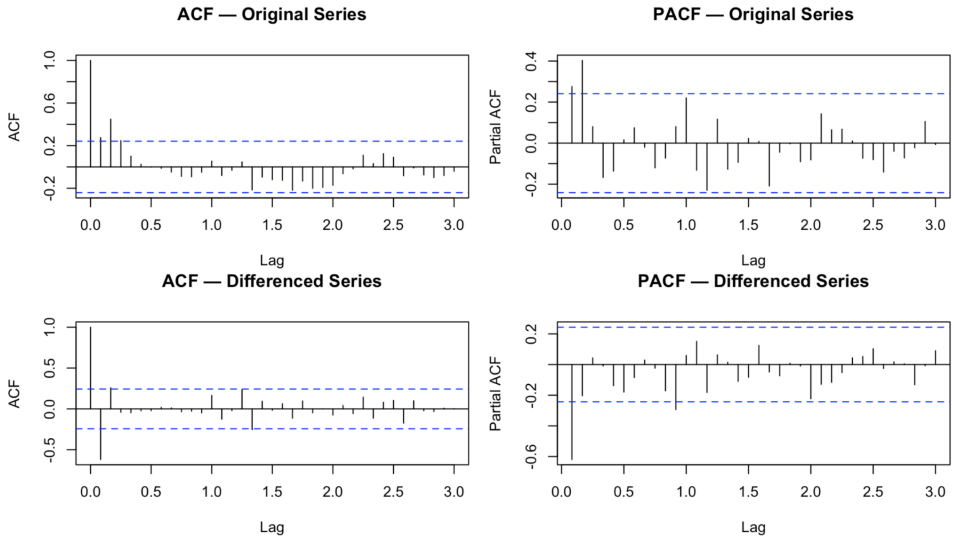

Monthly Tech Layoffs (2020–Present)
Bar = monthly total; Red line = raw trend; Dashed = LOESS smoother
Employees Laid Off
60,000
40,000
20,000
0
May 2020
Nov 2020
May 2021
Nov 2021
May 2022
Nov 2022
May 2023
Nov 2023
May 2024
Nov 2024
May 2025
Nov 2025
Month


Annual Tech Layoffs by Year
Total Employees Laid Off
150,000
100,000
50,000
0
61,529
6,790
127,577
176,946
122,661
86,202
2020
2021
2022
2023
2024
2025
Year


Heatmap of Monthly Layoffs by Year
Darker = more layoffs
Year
2025
2024
2023
2022
2021
2020
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
Layoffs
60,000
40,000
20,000

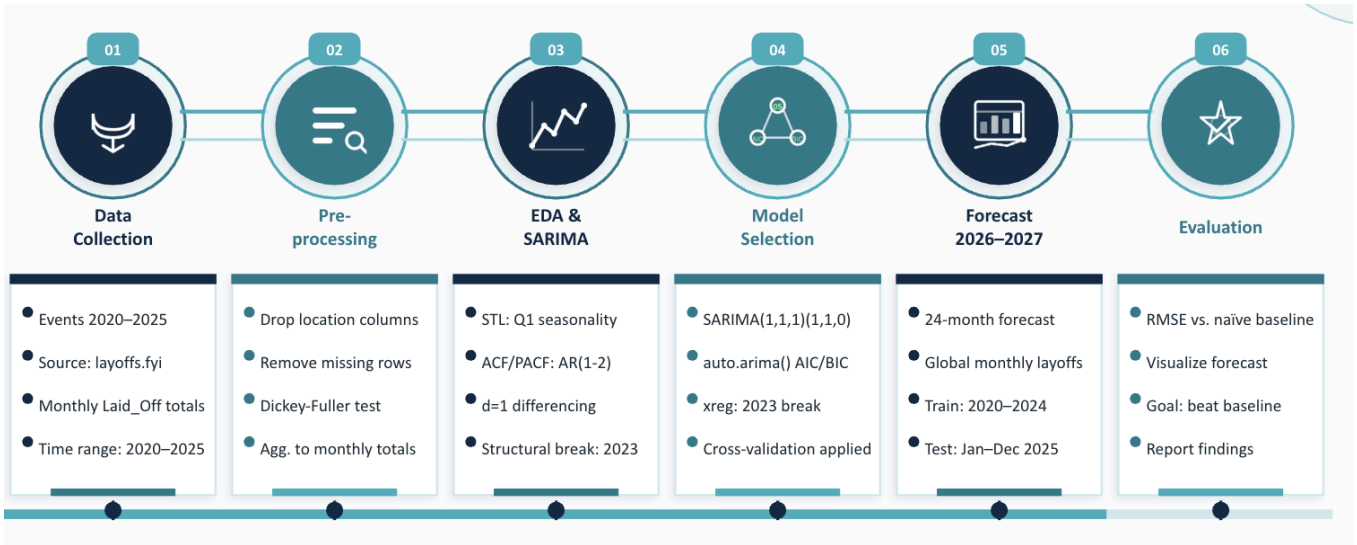


Series is non-stationary over time (spikes in 2023), which is why differencing is needed before fitting SARIMA. Seasonality in Jan-March, choosing Sarima over Arima.
Limitation: the remainder (noise) spikes dramatically around 2023 (cant be modeled seasonally since its a one time event).



ACF-OG: Autocorrelation decays slowly (non-stationary)
PACF-OG: Spike at lags 1 and 2 then cuts off quickly (strong first order autoregressive behavior) (this means that last months layoffs are the dominant predictor of this months)
ACF - differenced : Autocorrelation now drops sharply, lag 1 significant (differencing successfully removed the trend dependence) d=1
PACF- differenced: negative spike at lag 1 (potential indicator of over differencing) but since the ADF is stationary after differencing d=1 is still appropriate

Analysis Plan:



Preprocessing:
In order to clean and preprocess the Tech Layoff data, we will first drop any columns with unneeded data. This will include location-related columns such as longitude, latitude, region, etc. We will also drop any rows with missing data. After dropping unneeded columns and rows, we will aggregate the data into monthly layoff totals to create a time series appropriate for ARIMA/SARIMA model analysis. After cleaning the data, we will then perform a Dickey-Fuller Test to check if the data is stationary. If the data is not stationary, we will apply first-order differencing to transform the data. Finally, we will examine ACF/PACF plots to determine our model's parameters.

Methodology:
To analyze the data, we plan to use a SARIMA (Seasonal Autoregressive Integrated Moving Average) model to forecast global monthly tech layoffs for the remainder of 2026 and 2027. This model was chosen over an ARIMA model due to EDA revealing seasonality regarding layoffs. The dataset will be split into a training set (March 2020 through December 2024) and a held-out test set (January through December 2025, 12 months) to enable unbiased evaluation of forecast accuracy. For hyperparameter tuning, we will utilize auto.arima() from the Forecast R package. The function will iterate through different parameters to find the best ones, based on AIC and BIC model statistics. The EDA also revealed a structural break during the months of Janurary - March 2023; therefore, we will evaluate whether the inclusion of a binary external regressor improves out-of-sample performance. Cross-validation will also be applied during model training.

Evaluation:

To evaluate our model's performance, we will visualize our time series data and look at metrics such as RMSE (Root Mean Squared Error) and compare that with the error from a seasonal naive baseline. If the RMSE is less than the error, then we will know our model has succeeded.

Quantifiable Goal:
The goal of this project is to develop a time-series forecasting model to predict layoffs for the remainder of 2026 and 2027 using tech layoff data that achieves a prediction error (RMSE) on a held-out 12-month test set less than the error from a seasonal naive baseline.

Goal put in another way:
The goal of this project is to use tech layoff time series data to predict layoffs for the remainder of 2026 and 2027.